

NUMERICAL ANALYSIS

PROBLEM SET 4

ROB TOMPKINS

1. PROBLEM 1

1.a. We are deriving the 2-point open Newton-Coats formula

$$\int_a^b f(x)dx \approx I_{2,\text{open}}(f) = \frac{b-a}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right)$$

for evenly spaced points with $h = \frac{b-a}{n+2}$. Now we set $n = 2$ so that $h = \frac{b-a}{4}$, and we need to sort out

$$(1) \quad I_{2,\text{open}}(f) = \int_a^b \sum_{i=0}^2 L_{2,i}(x) f(x_i) dx.$$

But we need to shift to an easier or simpler variable to work with. So we set $x = a + t \cdot h$, with $dx = h \cdot dt$. Now as x moves from a to b we see that ($b = a + 4h$), t moves from 0 to 4, and we have

$$\int_a^b f(x)dx = h \int_0^4 f(a+ht)dt,$$

which we interpolate $f(a+ht)$ at $t = 1, 2, 3$. We write out the Lagrangian polynomials $L_{2,0}, L_{2,1}, L_{2,2}$ for the sake of the problem. This follows as

$$L_{2,0}(x) = \frac{(t-2)(t-3)}{(1-2)(1-3)} = \frac{(t-2)(t-3)}{2} = \frac{1}{2}(t^2 - 5t + 6)$$

$$L_{2,1}(x) = \frac{(t-1)(t-3)}{(2-1)(2-3)} = -(t-1)(t-3) = -t^2 + 4t - 3$$

$$L_{2,2}(x) = \frac{(t-1)(t-2)}{(3-1)(3-2)} = \frac{(t-1)(t-2)}{2} = \frac{1}{2}(t^2 - 3t + 2).$$

And we then need to integrate these Lagrange interpolants, which we do as

$$\begin{aligned}\int_0^4 \frac{1}{2}(t^2 - 5t + 6)dt &= \frac{t^3}{6} - \frac{5t^2}{4} + 3t \Big|_0^4 = \frac{8}{3} \\ \int_0^4 -t^2 + 4t - 3dt &= -\frac{t^3}{3} + 2t^2 - 3t \Big|_0^4 = -\frac{4}{3} \\ \int_0^4 \frac{1}{2}(t^2 - 3t + 2)dt &= \frac{t^3}{6} - \frac{3t^2}{2} + 2t \Big|_0^4 = \frac{8}{3}\end{aligned}$$

Thus, when we plug this into equation (1), we get the following

$$\begin{aligned}\int_a^b f(x)dx &\approx h \left(\frac{8}{3}f(a+h) - \frac{4}{3}f(a+2h) + \frac{8}{3}f(a+3h) \right) \\ &\approx \frac{4h}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right) \\ &\approx \frac{b-a}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right).\end{aligned}$$

which solves our problem

1.b. We must show that our quadrature is exact for $1, x, x^2, x^3$ but not for x^4 . Again we rescale the interval in teh fashion

$$x = a + ht, \quad h = \frac{b-a}{4}, x \in [a, b] \rightarrow t \in [0, 4]$$

So $Q(g) = \frac{4}{3} \left(2g(1) - g(2) - 2g(3) \right)$. But, we need to check if $I(g) = Q(g)$ for $g(t) \in \{1, x, x^2, x^3, x^4\}$. So we write

$$\begin{aligned}\int_0^4 1dt &= 4 \stackrel{?}{=} \frac{4}{3}(2 - 1 + 2) = 4 \quad \checkmark \\ \int_0^4 tdt &= \frac{t^2}{2} \Big|_0^4 = 8 \stackrel{?}{=} \frac{4}{3}(2 \cdot 1 - 2 + 2 \cdot 3) = 8 \quad \checkmark \\ \int_0^4 t^2dt &= \frac{t^3}{3} \Big|_0^4 = \frac{64}{3} \stackrel{?}{=} \frac{4}{3}(2 \cdot 1 - 4 + 2 \cdot 9) = \frac{4 \cdot 16}{3} = \frac{64}{3} \quad \checkmark \\ \int_0^4 t^3dt &= \frac{t^4}{4} \Big|_0^4 = 64 \stackrel{?}{=} \frac{4}{3}(2 \cdot 1 - 8 + 2 \cdot 27) = \frac{4 \cdot 48}{3} = \frac{192}{3} = 64 \quad \checkmark \\ \int_0^4 t^4dt &= \frac{t^5}{5} \Big|_0^4 = \frac{1024}{5} = 204.8 \stackrel{?}{=} \frac{4}{3}(2 - 32 + 2 \cdot 243) = \frac{1824}{3} = 608 \quad \times\end{aligned}$$

So we are done.

2. PROBLEM 2

Let

$$I(f) = \int_a^b f(x)dx$$

and let $[a, b]$ be divided into subintervals in the following fashion

$$\{[a, x_1 + h], [x_1 - h, x_1 + h], \dots, [x_n - h, b]\},$$

with $x_1, \dots, x_n$ being the midpoints of the subintervals. We thusly see that $h = \frac{b-a}{2n}$, that $x_j = a + (2j-1)h$, and that each of our subintervals has width $2h$. Now consider the j^{th} subinterval $[x_j - h, x_j + h]$. We can write $f(x)$ over the interval using Taylor's theorem as

$$f(x) = f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2,$$

for some η_x between x and x_j . So we integrate

$$\begin{aligned} \int_{x_j-h}^{x_j+h} f(x)dx &= \int_{x_j-h}^{x_j+h} \left(f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2 \right) dx \\ &= \int_{x_j-h}^{x_j+h} f(x_j)dx + \int_{x_j-h}^{x_j+h} f'(x_j)(x - x_j)dx + \int_{x_j-h}^{x_j+h} \frac{f''(\eta_x)}{2}(x - x_j)^2 dx \\ &= 2hf(x_j) + 0 + \frac{f''(\xi_j)}{2} \int_{x_j-h}^{x_j+h} (x - x_j)^2 dx, \end{aligned}$$

for some ξ_j clearly dependent upon our choice of η_x . But notice that our last integral can be manipulated by a transformation of variables in the following fashion

$$\int_{x_j-h}^{x_j+h} (x - x_j)^2 dx = \int_{-h}^h t^2 dt = \frac{2h^3}{3}.$$

Substituting that back in to the over all equation we see that

$$\int_{x_j-h}^{x_j+h} f(x)dx + \frac{f''(\xi_j)}{2} \frac{2h^3}{3}.$$

So we take this result and sum across all the subintervals to get that

$$\int_a^b f(x)dx = \sum_{j=1}^n \int_{x_j-h}^{x_j+h} f(x)dx = 2h \sum_{j=1}^n f(x_j) + \frac{h^3}{3} \sum_{j=1}^n f''(\xi_j).$$

But we notice that $\sum_{j=1}^n f''(\xi_j) = n f''(\xi)$ with $f''(\xi)$ representing the average of all $f''(\xi_j)$. And with some careful substitution, along with noticing that $n = \frac{b-a}{2h}$ by manipulating our definition of h , we see that

$$\int_a^b f(x)dx = 2h \sum_{j=1}^n f(x_j) + \frac{(b-a)h^2}{6} f''(\xi),$$

where $\xi \in [a, b]$, and we are done.

3. PROBLEM 3

We will determine the weights w_i and the quadrature points x_i in the three point Gauss-Legendre quadrature formula

$$\int_{-1}^1 f(x)dx \approx w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3).$$

Now by definition the degree of precision of this formula just falls out by us having to generate $2n - 1$ equations, so that we can use the method of undetermined coefficients for finding the weights. This yields a $2(3) - 1 = 5$ degree of precision. So we go through the process of generating the equations over which we'll solve for the weights by using the integrals of $1, x, x^2, x^3, x^4, x^5$, as

$$\begin{aligned} f(x) = 1 : \quad w_1 + w_2 + w_3 &= \int_{-1}^1 1 \cdot dx = 2, \\ f(x) = x : \quad w_1 x + w_2 x + w_3 x &= \int_{-1}^1 x dx = \frac{x^2}{2} \Big|_{-1}^1 = 0, \\ f(x) = x^2 : \quad w_1 x^2 + w_2 x^2 + w_3 x^2 &= \int_{-1}^1 x^2 dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{2}{3}, \\ f(x) = x^3 : \quad w_1 x^3 + w_2 x^3 + w_3 x^3 &= \int_{-1}^1 x^3 dx = \frac{x^4}{4} \Big|_{-1}^1 = 0, \\ f(x) = x^4 : \quad w_1 x^4 + w_2 x^4 + w_3 x^4 &= \int_{-1}^1 x^4 dx = \frac{x^5}{5} \Big|_{-1}^1 = \frac{2}{5}, \\ f(x) = x^5 : \quad w_1 x^5 + w_2 x^5 + w_3 x^5 &= \int_{-1}^1 x^5 dx = \frac{x^6}{6} \Big|_{-1}^1 = 0. \end{aligned}$$

Our goal now is to define a polynomial, call it $p(x)$, that vanishes at the quadrature points x_1, x_2, x_3 . So we have

$$\begin{aligned} p(x) &= (x - x_1)(x - x_2)(x - x_3) \\ &= x^3 - x_1 x^2 - x_2 x^2 - x_3 x^2 + x_1 x_2 x + x_1 x_3 x + x_2 x_3 x - x_1 x_2 x_3 \\ &= x^3 s_1 x^2 + s_2 x + s_3, \end{aligned}$$

where $s_1 = -(x_1 + x_2 + x_3)$, $s_2 = x_1 x_2 + x_1 x_3 + x_2 x_3$, and $s_3 = -x_1 x_2 x_3$.

Now to find our weights we set $p(x) = 0$, then $xp(x) = 0$, and finally $x^2p(x) = 0$. So let us begin with

$$\begin{aligned} p(x_i) = 0 &= x_i^3 + s_1x_i^2 + s_2x_i + s_3 = 0 \\ &= \sum w_ix_i^3 + s_1 \sum w_ix_i^2 + s_2 \sum w_ix_i + s_3 \sum w_i = 0 \\ &= 0 + s_1 \frac{2}{3} + 0 + s_3 \cdot (-2) \end{aligned}$$

So we have that

$$s_1 = -3s_3.$$

Now set $xp(x) = 0$, and let us see what we can find:

$$\begin{aligned} xp(x) &= x^4 + s_1x^3 + s_2x^2 + s_3x = 0 \\ &= \sum w_ix_i^4 + s_1 \sum w_ix_i^3 + s_2 \sum w_ix_i^2 + s_3 \sum w_ix_i = 0 \\ &= \frac{2}{5} + s_1 \cdot 0 + s_2 \frac{2}{3} + s_3 \cdot 0 = 0 \\ &= \frac{2}{5} + s_2 \frac{2}{3} = 0 \quad \Rightarrow \quad s_2 = -\frac{3}{5}. \end{aligned}$$

Next we set $x^2p(x) = 0$ to again have another equation over which we can solve for s_1, s_2, s_3 , which follows as

$$\begin{aligned} x^2p(x) &= x^5 + s_1x^4 + s_2x^3 + s_3x^2 = 0 \\ &= \sum w_ix_i^5 + s_1 \sum w_ix_i^4 + s_2 \sum w_ix_i^3 + s_3 \sum w_ix_i^2 = 0 \\ &= 0 + s_1 \frac{2}{5} + s_2 \cdot 0 + s_3 \frac{2}{3} = 0 \\ &= s_1 \frac{2}{5} + s_3 \frac{2}{3}. \end{aligned}$$

Because we have $s_1 = -3s_3$ we see that $-6s_3/5 + 2s_3/3 = s_3(2/3 - 6/5) = 0$. So, $s_3 = 0$ and thus $s_1 = 0$, which setting $p(x) = 0$ yields

$$p(x) = x^3 - \frac{3}{5}x = x \left(x^2 - \frac{3}{5} \right) = 0.$$

So we see that

$$x_1 = -\sqrt{\frac{3}{5}}, \quad x_2 = 0, \quad x_3 = \sqrt{\frac{3}{5}}.$$

So now we solve for the weights using our first three $f(x)$ equations from the beginning. We begin with,

$$(2) \quad \sum w_i = w_1 + w_2 + w_3 = 2.$$

Next, we see that

$$\sum w_i x_i = -w_1 \sqrt{\frac{3}{5}} + w_2 \cdot 0 + w_3 \sqrt{\frac{3}{5}} = 0 \quad \Longleftrightarrow \quad w_1 = w_3.$$

Lastly we compute,

$$\sum w_i x_i^2 = w_1 \frac{3}{5} + w_3 \frac{3}{5} = \frac{6w_1}{5} = \frac{2}{3} \quad \Longleftrightarrow \quad w_1 = w_3 = \frac{5}{9}$$

Which plugging back into equation (??) yields

$$\frac{5}{9} + w_2 + \frac{5}{9} = 2 \quad \Longleftrightarrow \quad w_2 = \frac{8}{9}.$$

So we see that

$$\int_{-1}^1 f(x) dx \approx \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right),$$

and we are done.

4. PROBLEM 4